

Name	K Dhyana samaga
Registration No.	261091010031
Subject	Principal of Secure Coding
	Submission - 1

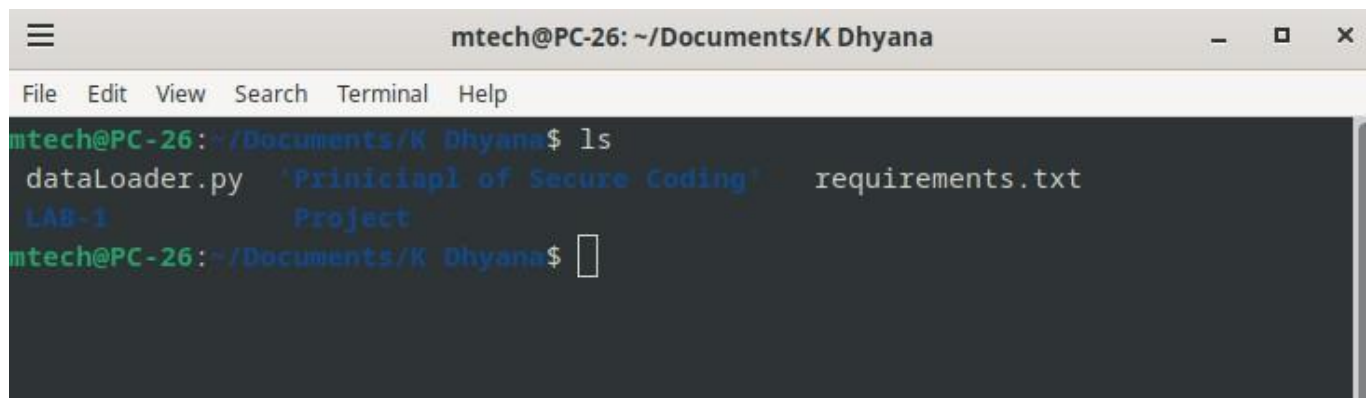
Linux commands

1. List the files present in the current folder

The ls command in Linux is used to list files and directories present in the current working directory.

```
ls
```

Output

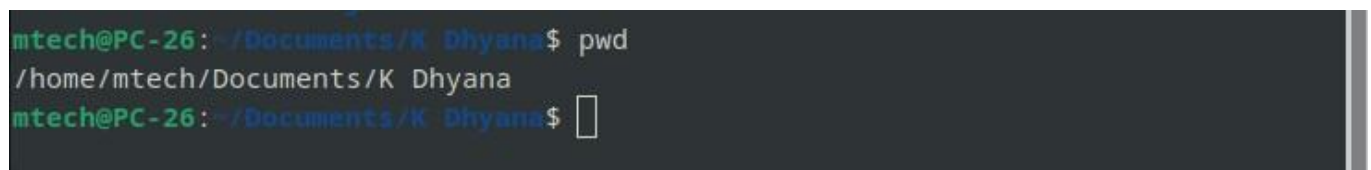
A terminal window titled 'mtech@PC-26: ~/Documents/K Dhyana' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'ls' being executed, resulting in the output: 'dataLoader.py', 'Priniciapl of Secure Coding', 'requirements.txt', 'LAB-1', and 'Project'. The prompt is 'mtech@PC-26: ~/Documents/K Dhyana\$'.

2. View the path of current folder

The pwd command in Linux is used to display the full path of the current directory.

```
pwd
```

Output

A terminal window titled 'mtech@PC-26: ~/Documents/K Dhyana' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command 'pwd' being executed, resulting in the output: '/home/mtech/Documents/K Dhyana'. The prompt is 'mtech@PC-26: ~/Documents/K Dhyana\$'.

3. Create new directory

The mkdir command in Linux is used to create new directories or folders.

```
mkdir name_of_folder
```

Output

```
mtech@PC-26:~/Documents/K Dhyana$ mkdir new_folder
mtech@PC-26:~/Documents/K Dhyana$ ls
dataLoader.py  new_folder  Project
LAB-1         'Priniciapl of Secure Coding'  requirements.txt
mtech@PC-26:~/Documents/K Dhyana$
```

4. Navigate to folder

The cd command in Linux is used to change the current working directory.

```
cd folder_name
```

Output

```
mtech@PC-26:~/Documents/K Dhyana$ cd new_folder/
mtech@PC-26:~/Documents/K Dhyana/new_folder$
```

5. Delete the folder

The rmdir command in Linux is used to delete empty directories.

```
rmdir folder_name
```

Output

```
mtech@PC-26:~/Documents/K Dhyana$ rmdir new_folder/
mtech@PC-26:~/Documents/K Dhyana$ ls
dataLoader.py  'Priniciapl of Secure Coding'  requirements.txt
LAB-1         Project
mtech@PC-26:~/Documents/K Dhyana$
```

6. Read the content in the file

The cat command in Linux is used to display file contents and combine multiple files.

```
cat file_name
```

Output

```
mtech@PC-26: ~/Documents/K Dhyana$ cat requirements.txt
#Machine-Learning
numpy
pandas
scikit-learn

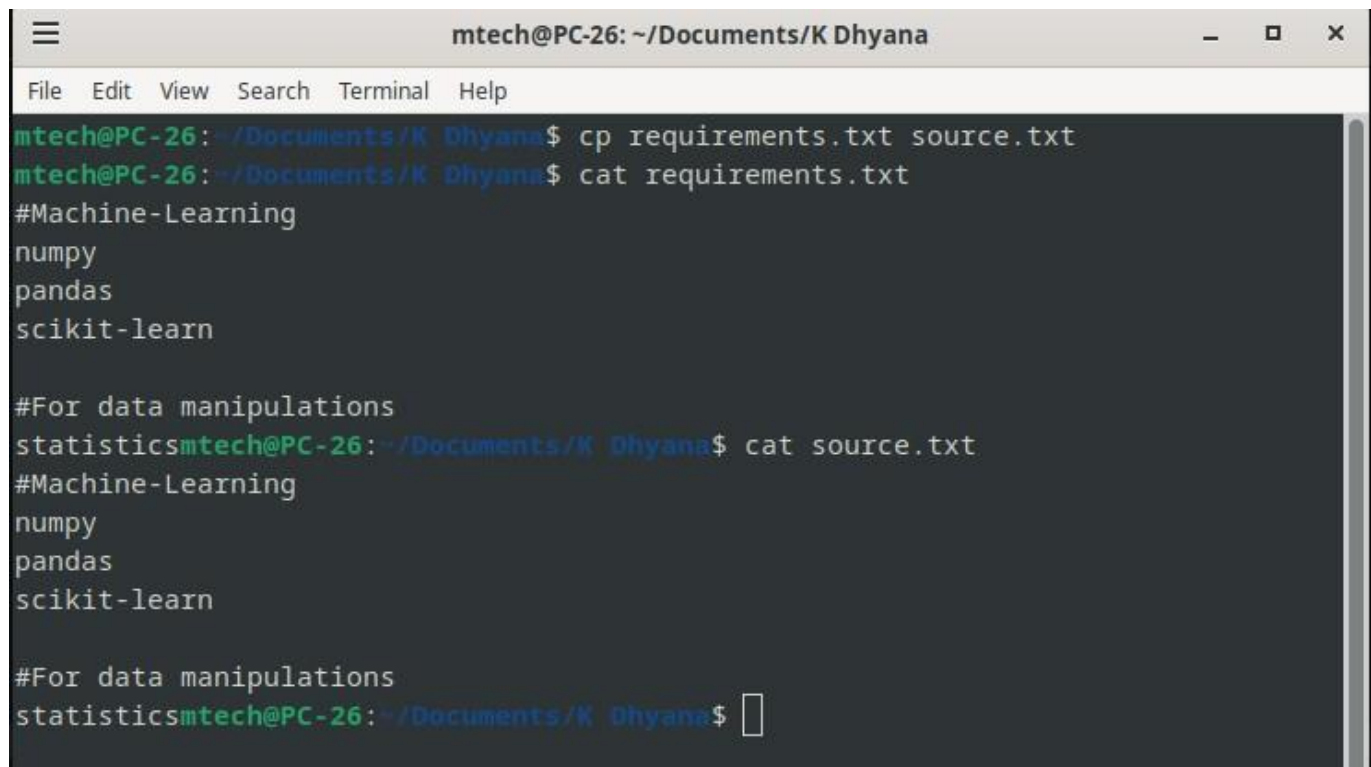
#For data manipulations
statisticsmtech@PC-26: ~/Documents/K Dhyana$
```

7. Copy the content from one file to another

The cp command in Linux is used to copy files or directories.

```
cp source destination
```

Output



```
mtech@PC-26: ~/Documents/K Dhyana
File Edit View Search Terminal Help
mtech@PC-26: ~/Documents/K Dhyana$ cp requirements.txt source.txt
mtech@PC-26: ~/Documents/K Dhyana$ cat requirements.txt
#Machine-Learning
numpy
pandas
scikit-learn

#For data manipulations
statisticsmtech@PC-26: ~/Documents/K Dhyana$ cat source.txt
#Machine-Learning
numpy
pandas
scikit-learn

#For data manipulations
statisticsmtech@PC-26: ~/Documents/K Dhyana$
```

8. Check today's date in terminal

The cal command in Linux is used to display the calendar.

```
cal
```

Output

```
statisticsmtech@PC-26: ~/Documents/K Dhyana$ cal
  August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
mtech@PC-26: ~/Documents/K Dhyana$
```

9. Check the system information

The uname command in Linux is used to display system information.

```
uname -a
```

Output

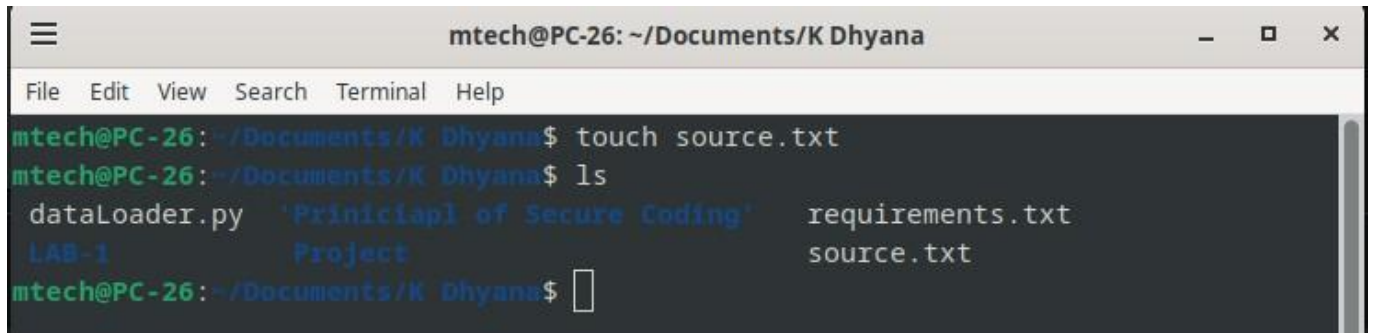
```
mtech@PC-26: ~/Documents/K Dhyana$ uname -a
Linux PC-26 6.1.0-52-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.180-1 (2026-08-03)
x86_64 GNU/Linux
mtech@PC-26: ~/Documents/K Dhyana$
```

10. Create the empty file

The touch command in Linux is used to create empty files and updates file timestamps.

```
touch file_name.extension
```

Output



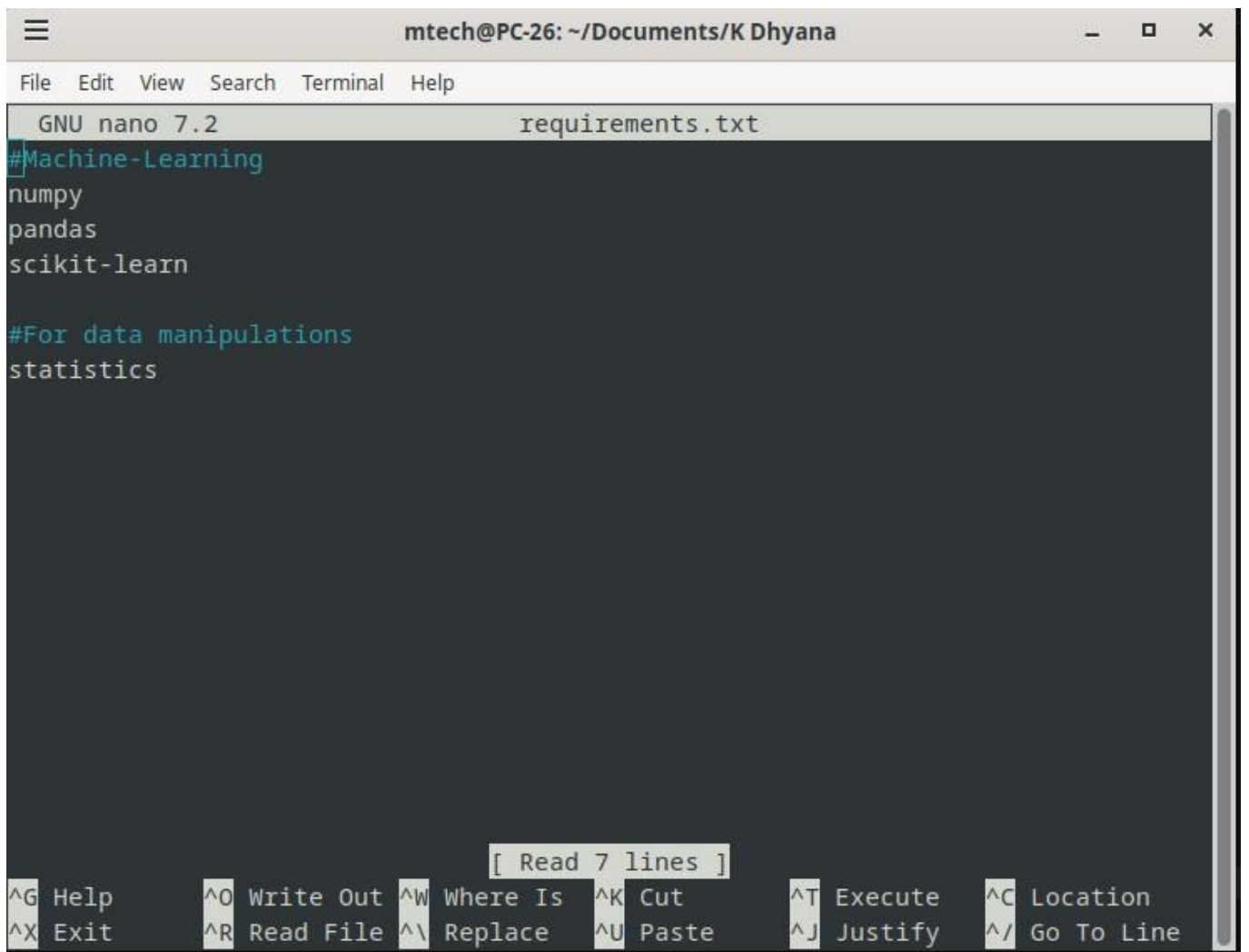
```
mtech@PC-26: ~/Documents/K Dhyana
File Edit View Search Terminal Help
mtech@PC-26:~/Documents/K Dhyana$ touch source.txt
mtech@PC-26:~/Documents/K Dhyana$ ls
dataLoader.py 'Priniciapl of Secure Coding' requirements.txt
LAB-1 Project source.txt
mtech@PC-26:~/Documents/K Dhyana$
```

11. Edit the file in nano editor.

Nano provides essential editing features, making it ideal for quick edits, creating configuration files, or writing scripts directly in the terminal.

```
nano file_name.extension
```

Output



```
mtech@PC-26: ~/Documents/K Dhyana
File Edit View Search Terminal Help
GNU nano 7.2 requirements.txt
#Machine-Learning
numpy
pandas
scikit-learn

#For data manipulations
statistics

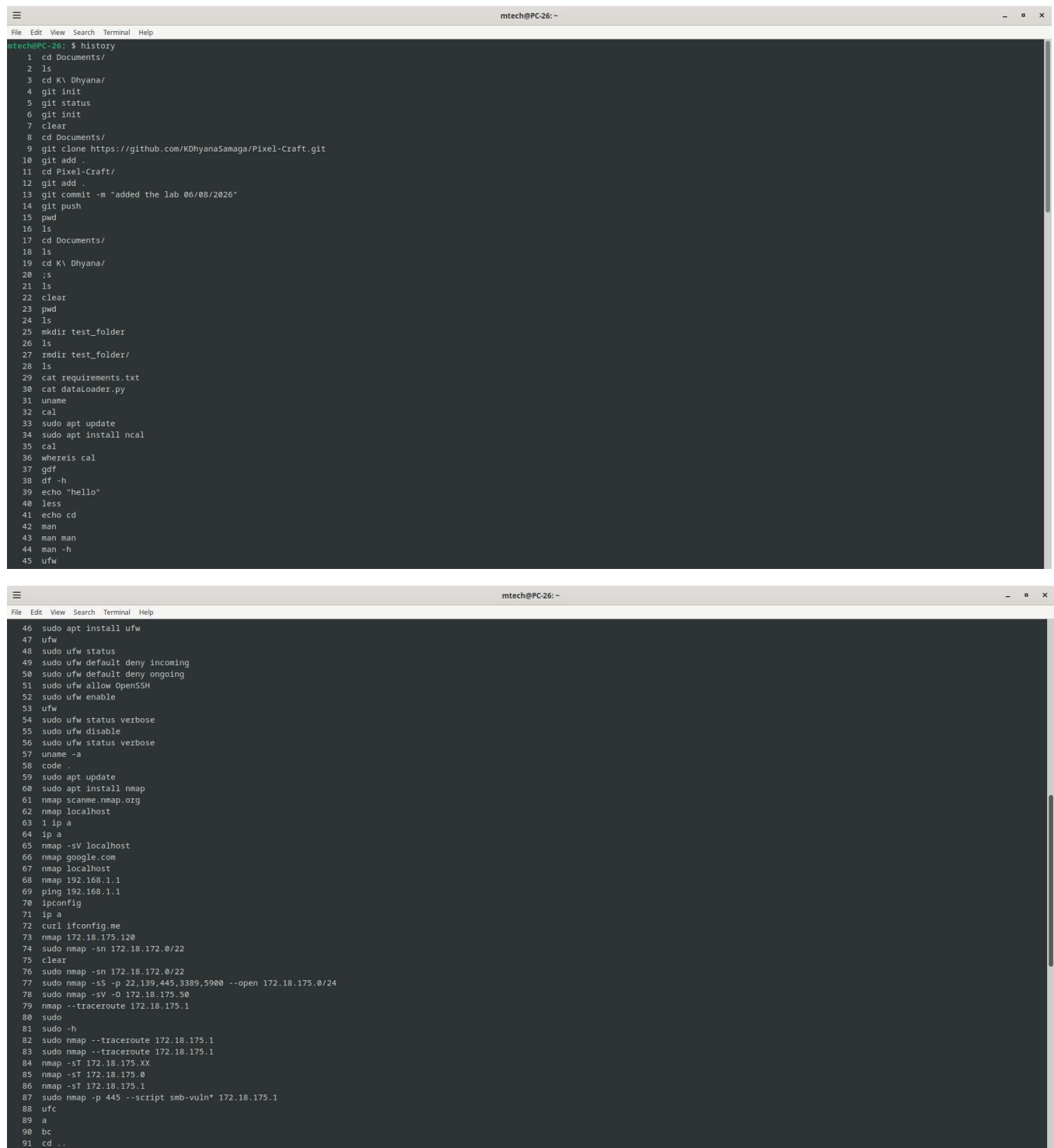
[ Read 7 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line
```

12. Shows a list of recently executed commands.

The history command in Linux displays a list of commands that were previously entered in the terminal.

history

Output



```
mtech@PC-26: ~  
File Edit View Search Terminal Help  
mtech@PC-26: $ history  
 1 cd Documents/  
 2 ls  
 3 cd K\ Dhyana/  
 4 git init  
 5 git status  
 6 git init  
 7 clear  
 8 cd Documents/  
 9 git clone https://github.com/KDhyanaSamaga/Pixel-Craft.git  
10 git add .  
11 cd Pixel-Craft/  
12 git add .  
13 git commit -m "added the lab 06/08/2026"  
14 git push  
15 pwd  
16 ls  
17 cd Documents/  
18 ls  
19 cd K\ Dhyana/  
20 ls  
21 ls  
22 clear  
23 pwd  
24 ls  
25 mkdir test_folder  
26 ls  
27 rmdir test_folder/  
28 ls  
29 cat requirements.txt  
30 cat dataLoader.py  
31 uname  
32 cal  
33 sudo apt update  
34 sudo apt install ncal  
35 cal  
36 whereis cal  
37 gdf  
38 df -h  
39 echo "hello"  
40 less  
41 echo cd  
42 man  
43 man man  
44 man -h  
45 ufw  
46 sudo apt install ufw  
47 ufw  
48 sudo ufw status  
49 sudo ufw default deny incoming  
50 sudo ufw default deny ongoing  
51 sudo ufw allow OpenSSH  
52 sudo ufw enable  
53 ufw  
54 sudo ufw status verbose  
55 sudo ufw disable  
56 sudo ufw status verbose  
57 uname -a  
58 code .  
59 sudo apt update  
60 sudo apt install nmap  
61 nmap scanme.nmap.org  
62 nmap localhost  
63 l ip a  
64 ip a  
65 nmap -sV localhost  
66 nmap google.com  
67 nmap localhost  
68 nmap 192.168.1.1  
69 ping 192.168.1.1  
70 ipconfig  
71 ip a  
72 curl ifconfig.me  
73 nmap 172.18.175.120  
74 sudo nmap -sn 172.18.172.0/22  
75 clear  
76 sudo nmap -sn 172.18.172.0/22  
77 sudo nmap -sS -p 22,139,445,3389,5900 --open 172.18.175.0/24  
78 sudo nmap -sV -O 172.18.175.50  
79 nmap --traceroute 172.18.175.1  
80 sudo  
81 sudo -h  
82 sudo nmap --traceroute 172.18.175.1  
83 sudo nmap --traceroute 172.18.175.1  
84 nmap -sT 172.18.175.XX  
85 nmap -sT 172.18.175.0  
86 nmap -sT 172.18.175.1  
87 sudo nmap -p 445 --script smb-vuln* 172.18.175.1  
88 ufc  
89 a  
90 bc  
91 cd ..
```

```
mtech@PC-26: ~  
File Edit View Search Terminal Help  
94 2*5  
95 bc 2*5  
96 echo "2*5" | bc  
97 echo "2*6*10" | bc  
98 clear  
99 cd Desktop/  
100 ls  
101 CLE  
102 clear  
103 cd Docu  
104 cd ..  
105 cd Documents/  
106 clear  
107 ls  
108 cd K\ Dhyana/  
109 cd K\ Dhyana/  
110 clear  
111 ls  
112 pwd  
113 ls  
114 mkdir new_folder  
115 ls  
116 cd new_folder/  
117 cd ..  
118 mkdir new_folder/  
119 ls  
120 cat requirements.txt  
121 clear  
122 touch source.txt  
123 ls  
124 touch destination  
125 ls  
126 rm destination  
127 ls  
128 cp requirements.txt source.txt  
129 cat requirements.txt  
130 clear  
131 cp requirements.txt source.txt  
132 cat requirements.txt  
133 cat source.txt  
134 cal  
135 uname -a  
136 clear  
137 nano requirements.txt  
138 history  
mtech@PC-26: $
```

13. Search files

The locate command in Linux is used to find files.

```
find . -name "file_name"
```

Output

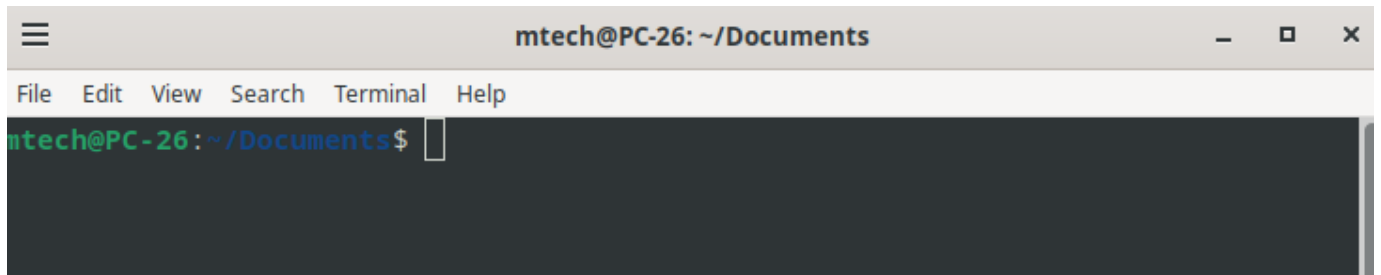
```
mtech@PC-26: ~/Documents  
File Edit View Search Terminal Help  
mtech@PC-26:~/Documents$ find . -name "requirements.txt"  
./K Dhyana/requirements.txt  
mtech@PC-26:~/Documents$
```

14. Clear the terminal screen

The clear command in Linux is used to clear the terminal screen.

```
clear
```

Output

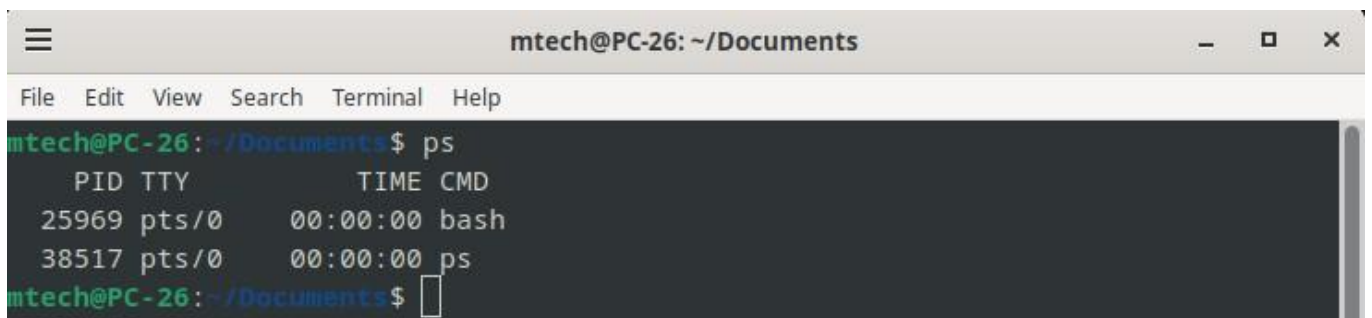


15. Display running processes.

The ps command in Linux is used to display running processes.

```
ps
```

Output



16. Display command manuals

The man command in Linux is used to display command manuals.

```
man command_name
```

Output


```
mtech@PC-26: ~/Documents
File Edit View Search Terminal Help
mtech@PC-26:~/Documents$ man ls
```

```
mtech@PC-26: ~/Documents
File Edit View Search Terminal Help
LS(1)                                User Commands                                LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default).
    Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
    fied.

    Mandatory arguments to long options are mandatory for short options
    too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

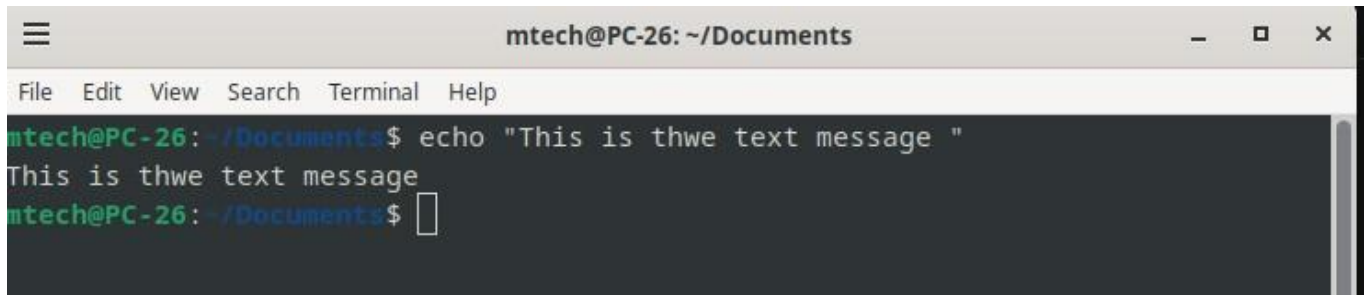
    --author
    Manual page ls(1) line 1 (press h for help or q to quit)
```

17. Display text in the terminal.

The echo command in Linux is used to display text in the terminal.

```
echo "Display text"
```

Output

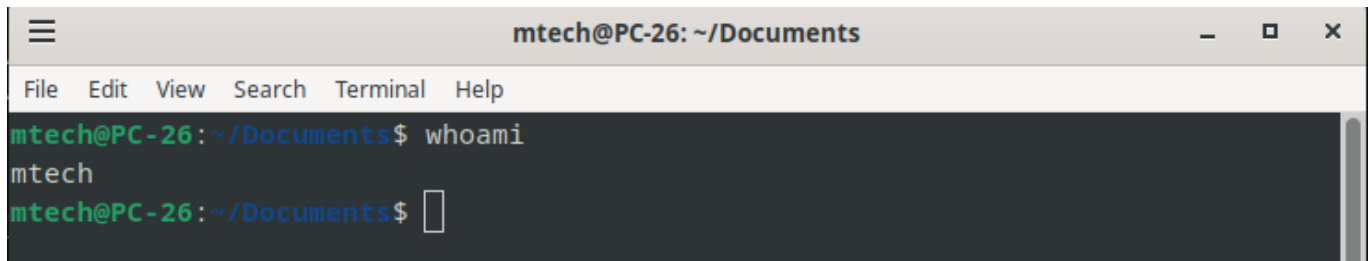
A terminal window titled 'mtech@PC-26: ~/Documents' with a menu bar (File, Edit, View, Search, Terminal, Help). The prompt is 'mtech@PC-26: ~/Documents\$'. The user enters 'echo "This is thwe text message "' and the output is 'This is thwe text message'. The prompt returns to 'mtech@PC-26: ~/Documents\$' with a cursor.

```
mtech@PC-26: ~/Documents$ echo "This is thwe text message "
This is thwe text message
mtech@PC-26: ~/Documents$
```

18. Display the current user

The whoami command in Linux is used to display the current user.

```
whoami
```

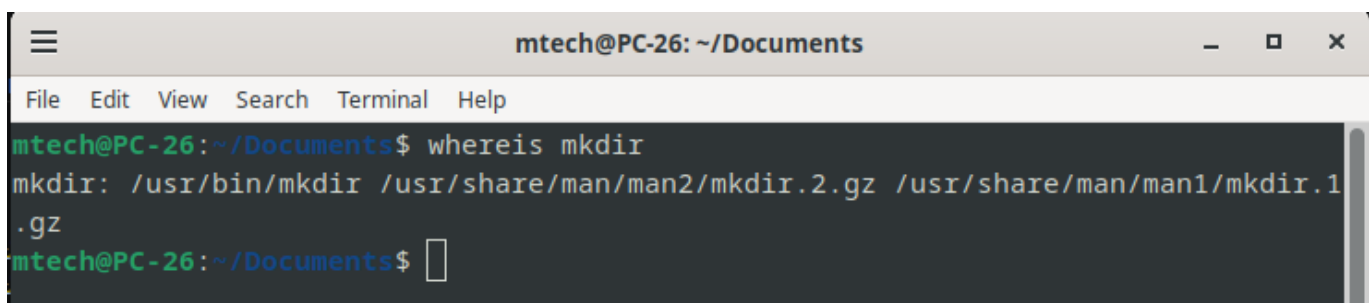
A terminal window titled 'mtech@PC-26: ~/Documents' with a menu bar (File, Edit, View, Search, Terminal, Help). The prompt is 'mtech@PC-26: ~/Documents\$'. The user enters 'whoami' and the output is 'mtech'. The prompt returns to 'mtech@PC-26: ~/Documents\$' with a cursor.

```
mtech@PC-26: ~/Documents$ whoami
mtech
mtech@PC-26: ~/Documents$
```

19. Locate command files

The whereis command in Linux is used to locate command files.

```
whereis command_name
```

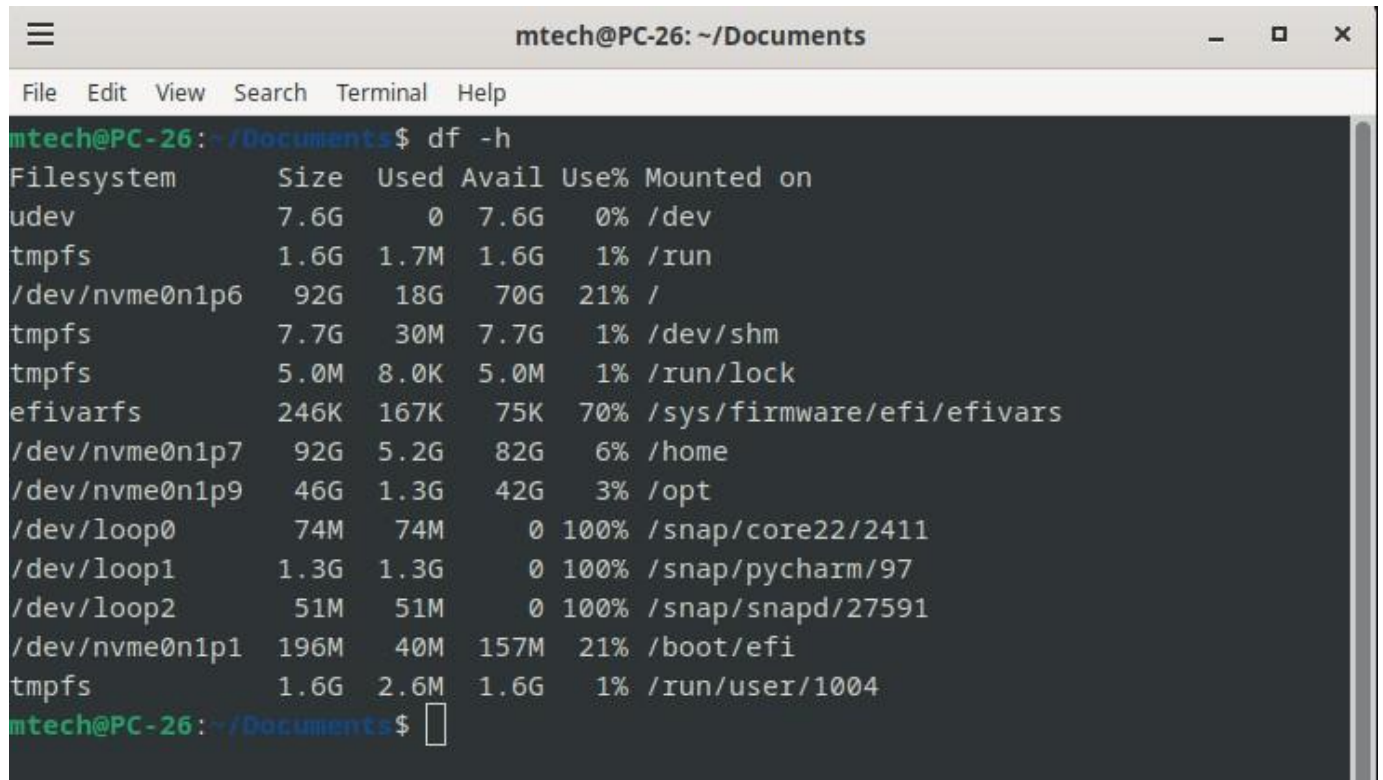
A terminal window titled 'mtech@PC-26: ~/Documents' with a menu bar (File, Edit, View, Search, Terminal, Help). The prompt is 'mtech@PC-26: ~/Documents\$'. The user enters 'whereis mkdir' and the output is 'mkdir: /usr/bin/mkdir /usr/share/man/man2/mkdir.2.gz /usr/share/man/man1/mkdir.1.gz'. The prompt returns to 'mtech@PC-26: ~/Documents\$' with a cursor.

```
mtech@PC-26: ~/Documents$ whereis mkdir
mkdir: /usr/bin/mkdir /usr/share/man/man2/mkdir.2.gz /usr/share/man/man1/mkdir.1
.gz
mtech@PC-26: ~/Documents$
```

20. Display disk space usage

The df command in Linux is used to display disk space usage.

```
df -h
```



A terminal window titled "mtech@PC-26: ~/Documents" with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal displays the output of the command `df -h`. The output is a table showing disk usage for various filesystems. The prompt `mtech@PC-26: ~/Documents$` is followed by a cursor.

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	7.6G	0	7.6G	0%	/dev
tmpfs	1.6G	1.7M	1.6G	1%	/run
/dev/nvme0n1p6	92G	18G	70G	21%	/
tmpfs	7.7G	30M	7.7G	1%	/dev/shm
tmpfs	5.0M	8.0K	5.0M	1%	/run/lock
efivarfs	246K	167K	75K	70%	/sys/firmware/efi/efivars
/dev/nvme0n1p7	92G	5.2G	82G	6%	/home
/dev/nvme0n1p9	46G	1.3G	42G	3%	/opt
/dev/loop0	74M	74M	0	100%	/snap/core22/2411
/dev/loop1	1.3G	1.3G	0	100%	/snap/pycharm/97
/dev/loop2	51M	51M	0	100%	/snap/snapd/27591
/dev/nvme0n1p1	196M	40M	157M	21%	/boot/efi
tmpfs	1.6G	2.6M	1.6G	1%	/run/user/1004

```
mtech@PC-26: ~/Documents$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            7.6G   0    7.6G   0% /dev
tmpfs           1.6G  1.7M   1.6G   1% /run
/dev/nvme0n1p6  92G   18G   70G   21% /
tmpfs           7.7G   30M   7.7G   1% /dev/shm
tmpfs           5.0M   8.0K   5.0M   1% /run/lock
efivarfs        246K  167K   75K   70% /sys/firmware/efi/efivars
/dev/nvme0n1p7  92G   5.2G   82G    6% /home
/dev/nvme0n1p9  46G   1.3G   42G    3% /opt
/dev/loop0      74M   74M    0  100% /snap/core22/2411
/dev/loop1     1.3G   1.3G    0  100% /snap/pycharm/97
/dev/loop2      51M   51M    0  100% /snap/snapd/27591
/dev/nvme0n1p1 196M   40M  157M   21% /boot/efi
tmpfs           1.6G   2.6M   1.6G   1% /run/user/1004
mtech@PC-26: ~/Documents$
```